LESSON 11.1
REPRESENTING BIVARIATE DATA

LESSON INTRODUCTION

MA.8.DP.1.1 Given a set of real-world bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.

LEARNING TARGETS

- I can determine the independent and dependent variables in a given set of real-world bivariate data.
- I can determine if the data is best represented by a line graph or scatter plot.
- I can determine an appropriate scale for the axis representing each variable.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- What is bivariate data?
- How can bivariate data be represented?
- What are the benefits and drawbacks of using a scatter plot?
- What are the benefits and drawbacks of using a line graph?

LESSON NARRATIVE

This lesson provides the underpinnings for students' learning about bivariate data. Students learn that bivariate data expresses two characteristics of a data set and examine ways in which it can be represented. To that end, students consider scatter plots with a line of fit and line graphs as possibilities. Students also examine how scale can impact the appearance and interpretation of data. This lesson then progresses to creating scatter plots and line graphs.

COMPONENT SUMMARY

11.1.1 WARM-UP *(~5 MIN.)*

Career exploration

11.1.2 GUIDED INSTRUCTION *(15-20 MIN.)*

Represent bivariate data

11.1.3 GUIDED PRACTICE *(10 MIN.)*

Analyze bivariate data

11.1.4 COLLABORATION *(10 MIN.)*

Choose an appropriate graph

11.1.5 WRAP-UP *(~5 MIN.)*

Reflection

RESOURCES

GLOSSARY

Key Terms:

- Bivariate data
- Scatter plot
- Line of fit
- Line graph

Additional Terms: N/A

MATERIALS

- (Optional) Examples of scatter plots without lines of fit
- (Optional) Blank pieces of paper

ADDITIONAL PREPARATION

- Warm-Up 11.1.1 contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think a line of fit represents specific points of data.
- Students may think the x- and y-axes of a graph must have the same scale.

THINGS TO CONSIDER

- Students can compare the data on the Guided Instruction scatter plot and line graph, as well as the table from which the graphs are created. Have students note the similar points in all three sources and observe how the line of fit provides a trend line through the data, but does not add data points.
- Guide students to describe the scales used on the graphs in the lesson. Some students may also benefit from writing the scale beside each axis.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

The lesson invites students to compare, contrast, and connect the same data represented in different ways. Demonstrate your thinking out loud about how each type of graph represents the same data. Then encourage students to reflect on the process, using your example as a bridge to their own thinking.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Students will be working to discern the differences between different graphical representations of the same data set. Clarifications within this benchmark emphasize choosing an appropriate representation based on a context or purpose. Encourage students to consider how different representations may be beneficial in different contexts.

DIFFERENTIATE INSTRUCTION

Encourage students to create a Frayer model to help them remember the vocabulary in the lesson. Provide larger versions of the graphs for students with visual challenges. Allow students to respond in multiple ways including in writing or verbal discussion. Provide multiple opportunities for students to discuss and listen to the concepts presented in the lesson.

11.1.1 WARM-UP: ARMY DENTIST

ENRICHMENT

- Were there any classes that Captain Romero recommended for future dentists that surprised you?
- What topics in math classes might be applicable to dentistry?

11.1.1 Warm-Up: Army Dentist

When Captain Ryan Romero was asked how he got started as an Army dentist, he said his first job was to be a dental assistant. He was sitting on the other side of the chair helping the dentist and he enjoyed it so much that he decided to pursue a career in Army dentistry.

1. When thinking about your future career, it's helpful to explore areas of interest to help you identify careers you might like. What is a career you think you would like to have?
   Answers will vary.
   Sample response: One career that I'm interested in is being an archaeologist.

2. What could you do now or in the future to help you determine if that is a career you would enjoy?
   Answers will vary.
   Sample response: I could get an internship in the field to see if I would enjoy the work that archaeologists do.

11.1.2 GUIDED INSTRUCTION: REPRESENTING BIVARIATE DATA

11.1.2 Guided Instruction: Representing Bivariate Data

1. The U.S. Centers for Disease Control and Prevention (CDC) is the nation's health protection agency. Epidemiologists at the CDC study the spread and impact of diseases. The table shows the number of symptomatic cases of influenza, also called the flu, in millions, in the U.S. since 2011.

Years Since 2011	0	1	2	3	4	5	6	7	8	9
Number of Cases, in millions	21	9.3	34	30	30	24	29	45	36	38

   a. What year does the number 6 represent?
   2017

   b. Keanu and Francisca were discussing what they noticed about the values in the table.

Keanu	Francisca
I noticed the actual year was not used in the table. I think this makes sense because the flu may not have existed in 1700.	I noticed the information given in the table stated the information was from a study. I think the study did not start until 2011.

   Discuss Keanu's and Francisca's ideas with your partner.
   Answers will vary.

2. Discuss with your partner what you notice in the data. Summarize your discussion.
   Answers will vary.
   Sample response: I notice that in 2012, the number of flu cases dropped, but then it increased in 2013. The number of flu cases was greatest in 2018.

This data set is an example of ***bivariate data*** because two variables are being compared.

> ***Bivariate data*** is data that measures two characteristics of a population.

ENRICHMENT

Help students make sense of the table through questioning:

- How is the table constructed?
- What is Year 0?
- How many flu cases occurred in that year?
- What trends do you notice in the number of flu cases? What may explain these trends?

TEACHER MOVES

Students may discuss why the table does not use specific years, and instead dates years since 2011. Encourage students to explore reasons why this may occur, such as the date when they started tracking that specific data, how scientists may want to see how the data varies over specific years, and/or how the years correspond to the years in a study.

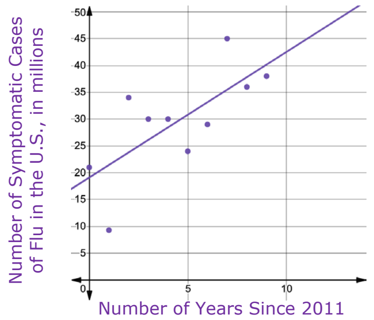
3. Based on the definition of bivariate data, what are the two characteristics measured in the table?
   The number of years since 2011 and the number of flu cases (in millions)

Bivariate numerical data can be represented in different ways. One way to represent the data is a ***scatter plot*** with a ***line of fit***.

> A ***scatter plot*** is a graph in the coordinate plane representing a set of bivariate numerical data that is used to observe the relationship between two variables.

> A ***line of fit*** is a line drawn on a scatter plot to estimate the relationship between two sets of data. A line of fit is also known as a trend line.

A scatter plot of the data on symptomatic cases of the flu in the U.S. is shown. The line of fit, $y = 2.34x + 19.1$, is included on the scatter plot, where y is the number of symptomatic cases of the flu in the U.S., in millions, and x is the number of years since 2011.



DIFFERENTIATE INSTRUCTION

To help students keep track of the vocabulary in the lesson, have students fold a piece of paper into quarters. Write a vocabulary word in each quarter along with its definition and a visual depiction of it.

TEACHER MOVES

Some students may think that a line of fit contains actual data points, instead of just showing a trend. Provide students with several scatter plots without a line of fit. Encourage students to orally describe the trend of the data. Then ask students to draw a line of fit to represent their verbal discussion.

11.1.2 GUIDED INSTRUCTION: REPRESENTING BIVARIATE DATA (CONTINUED)

DIFFERENTIATE INSTRUCTION

Create a large graph on the classroom floor; have students act as data points to create a scatter plot and use a piece of yarn for the line of fit. To recreate a line graph, have students hold a piece of yarn between them on the points.

ENRICHMENT

Encourage synthesis through intentional questioning:

- How does the line of fit show the trend of the scatter plot?
- How well does it help you better understand the data on the graph?
- How does the line graph show the connection between the points?

TEACHER MOVES

To further explore independent and dependent variables, encourage students to name pairs of data that are dependent, such as time and distance walked, or items purchased and amount spent. Have students identify the independent and dependent variable in each pair, and explain why it is independent or dependent.

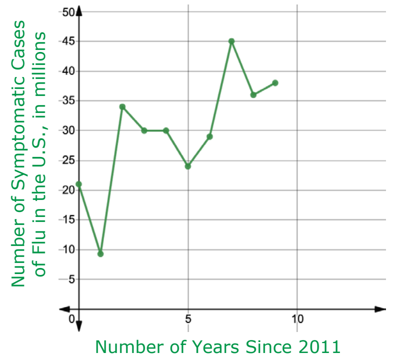
ENRICHMENT

- Why is the number of years the independent variable? How do you know that?
- Why is the number of cases of the flu the dependent variable? How do you know that?

The same data can be represented using a ***line graph***.

A ***line graph*** is a graph that displays numerical data using connected line segments.

A line graph of the data on symptomatic cases of the flu in the U.S. is shown.



4. Discuss with your partner the similarities and differences between the scatter plot and the line graph.
   Answers will vary.
   a. Complete the table.

Similarities Between Scatter Plots and Line Graphs	Differences Between Scatter Plots and Line Graphs
Answers will vary. The axes have the same labels and scales. Both have the points plotted that represent each data point.	*Answers will vary. In a line graph, the points are connected, but they are not in a scatter plot.*

   b. Mateo is interested in determining how the number of flu cases in the U.S. has changed each year since 2011. Explain which graph is best for him to use.
   Answers will vary.
   Mateo should use the line graph because he is looking at the change for each individual year, and the line segments in between each point show the change from year to year.

   c. Londyn is interested in the overall trend in the number of flu cases in the U.S. since 2011. Explain which graph is best for her to use.
   Answers will vary.
   Londyn should use the scatter plot to see the overall trend because the line of fit shows the trend over time.

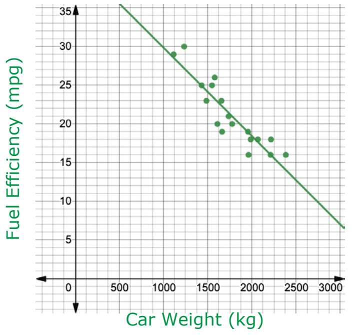
   d. Discuss with your partner which variable is the independent variable and which is the dependent variable in this context. Identify your choices.

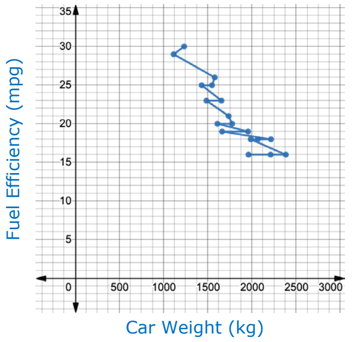
Independent Variable	○ Number of cases of the flu ○ *Years since 2011*
Dependent Variable	○ *Number of cases of the flu* ○ Years since 2011

Both scatter plots and line graphs can be used to observe the relationship between two variables in a set of bivariate data. While overall trends can be seen in both graphs, line graphs allow for closer analysis from one point to another in the data set.

11.1.3 Guided Practice: Representing Bivariate Data

1. A scatter plot and a line graph comparing the weights, in kilograms (kg), and fuel efficiencies, in miles per gallon (mpg), of 18 cars are shown.





a. Discuss with your partner which graph you think more clearly shows the relationship between a car's weight and its fuel efficiency. Summarize your discussion.
Answers will vary.
We think the scatterplot more clearly shows the relationship between a car's weight and its fuel efficiency because it shows that there is an overall trend. The line graph is hard to follow.

b. Complete the statement to describe the relationship shown on the graph identified in Part A.

The graph shows that, in general, a car's fuel efficiency [○ increases / ○ decreases] as the car's weight increases.

c. Kate noticed that there were cars of two different weights with the same fuel efficiency of 25 mpg. How does this impact the visual representation of the data on the line graph?
The points representing the two cars of different weights that both had the fuel efficiency of 25 mpg are connected by a horizontal line on the line graph.

When the goal of analyzing a set of bivariate numerical data is to observe the relationship between the two variables overall, a scatter plot is most useful.

11.1.3 GUIDED PRACTICE: REPRESENTING BIVARIATE DATA

TEACHER MOVES

MA.K12.MTR.2.1: Multiple Representations

Encourage students to examine how the same data set can be modeled in multiple ways. Show students that different representations can have different purposes and can be useful in different situations. Ask students what criteria they could use to choose a representation based on a context or purpose.

DIFFERENTIATE INSTRUCTION

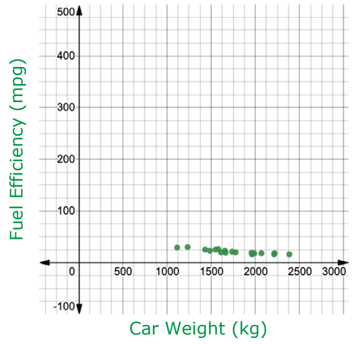
To allow students to have more opportunities to discuss their observations and more easily attend to others, combine two pairs into a small group of four.

11.1.3 GUIDED PRACTICE: REPRESENTING BIVARIATE DATA (CONTINUED)

ENRICHMENT

Encourage students to analyze the graph. To help students determine the impact of the scale, ask questions like: What is the scale of the x-axis? What is the scale of the y-axis? How would the data appear if the scale of the x-axis had smaller intervals? Larger intervals? How would the data appear if the scale of the y-axis had smaller intervals? Larger intervals?

d. A scatter plot representing the same bivariate data set, but using a different scale is shown.



Explain how the change in scale could influence how the relationship between the two variables is interpreted.

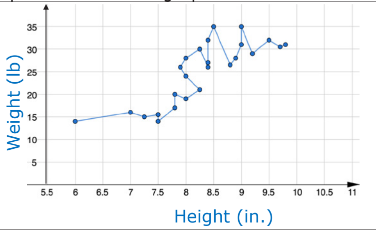
Answers will vary.

The scale could influence how steep the line of fit appears on the graph, which could impact how much someone thinks that the dependent variable changes as the independent variable changes. On this graph, it doesn't seem like there is much change the fuel efficiency for cars of different weights because the line of best fit would not be very steep.

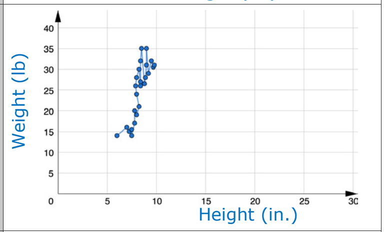
11.1.4 Collaboration: Choosing an Appropriate Graph

1. The relationship between the height, in inches (in.), and weight, in pounds (lb), of 25 dachshunds, a breed of dogs, is represented on the graphs.

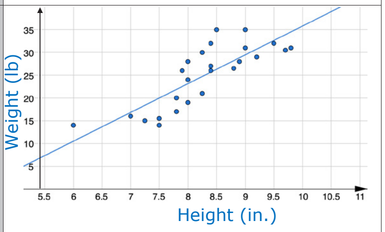
Line Graph A



Line Graph B



Scatter Plot A



11.1.4 COLLABORATION: CHOOSING AN APPROPRIATE GRAPH

TEACHER MOVES

Compare and Connect

Encourage students to examine the similarities and differences in how the same data is presented on different types of graphs and using different scales. Some students may benefit from creating a t-chart of the pros and cons of using a specific graph to represent a data set.

DIFFERENTIATE INSTRUCTION

Provide larger graphs for students with visual challenges. Trace the lines in bright colors and highlight the numbers indicating the scale.

11.1.4 COLLABORATION: CHOOSING AN APPROPRIATE GRAPH (CONTINUED)

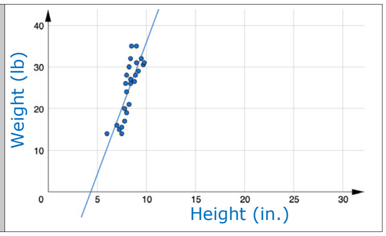
ENRICHMENT

Prompt students to make the connection between this question and the Pythagorean theorem, which we have been working on in the last few lessons.

TEACHER MOVES

Consider asking students to identify the independent and dependent variables in order to reinforce those concepts.

Scatter Plot B

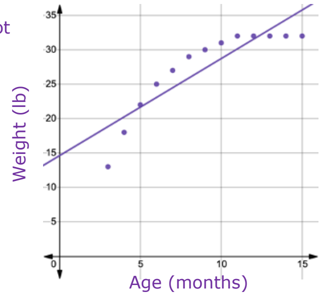


a. Work with your partner to explain which of the graphs is best to show the relationship between the weight and height of these dogs. Be sure to include information about the type of graph and the scale in your explanation.
Answers will vary.
Scatterplot A is best to show the relationship. A scatterplot works best in this case because it more clearly shows the overall trend in the relationship between weight and height of the dachshunds. The scale of Scatterplot A is better than the scale of Scatterplot B because you can more easily see the individual points.

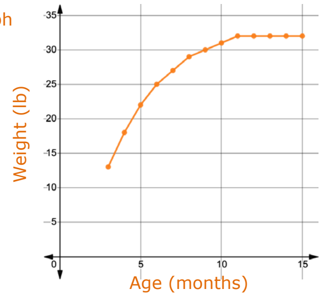
b. Describe the relationship between a dachshund's height and its weight using the graph identified in Part A.
Answers will vary.
The slope of the line of fit is positive, which means the dachshunds that are taller tend to weigh more as well.

2. The relationship between the age of a dachshund, in months, and its weight, in pounds (lb), is shown on the graphs.

Scatter Plot



Line Graph



Explain which of the graphs best demonstrates the relationship between the age and weight of dachshunds.

Answers will vary.
The line graph best shows the relationship because after a certain age, the weight stays the same instead of continuing to increase.

11.1.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Use students' responses to plan future lessons and learning experiences. Consider creating a small group for reteaching students who need help with more than one learning target. Think about pairing students who need help with one learning target with a student who feels that they could help someone. Also use students' responses to determine who could progress more quickly through future lessons.